

Genetic Parameters and Genome-wide Association Studies of Disease in Growing Pigs under a Polymicrobial Natural Disease Challenge

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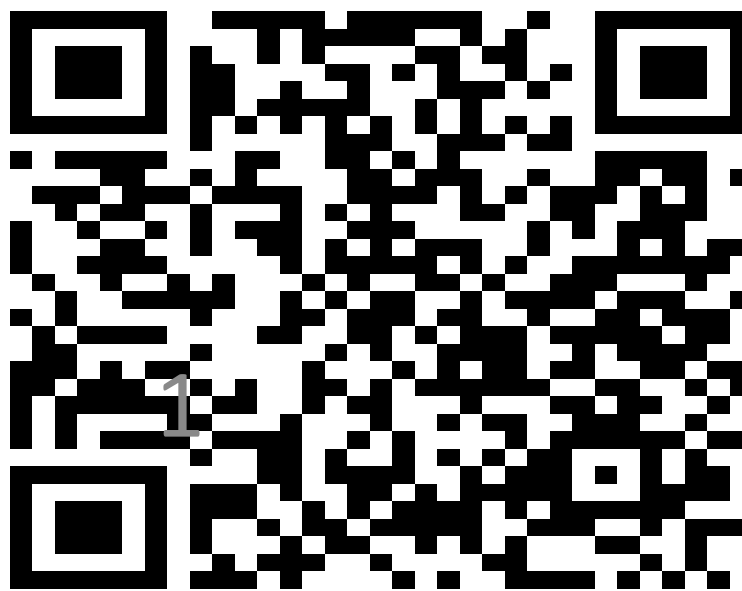
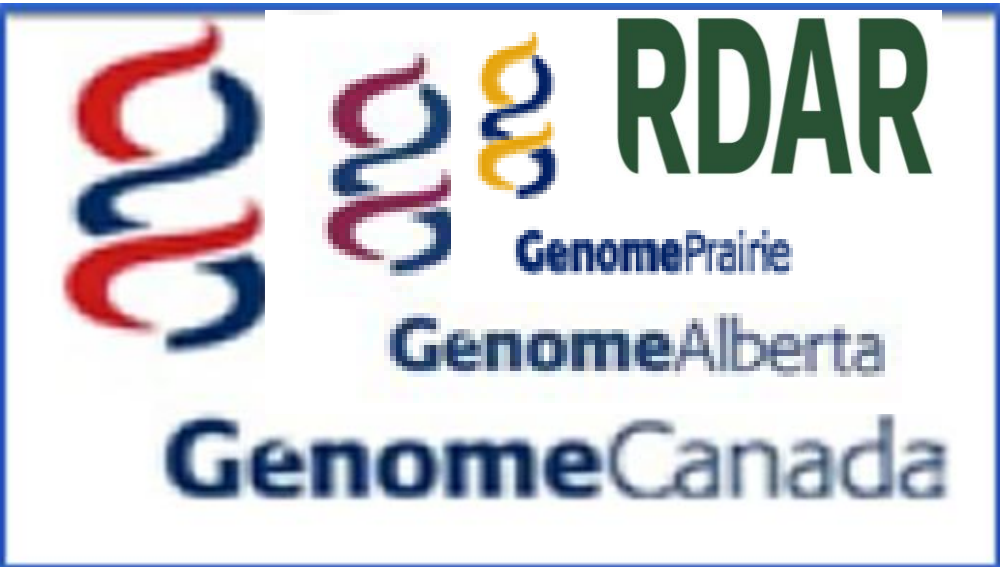
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Introduction

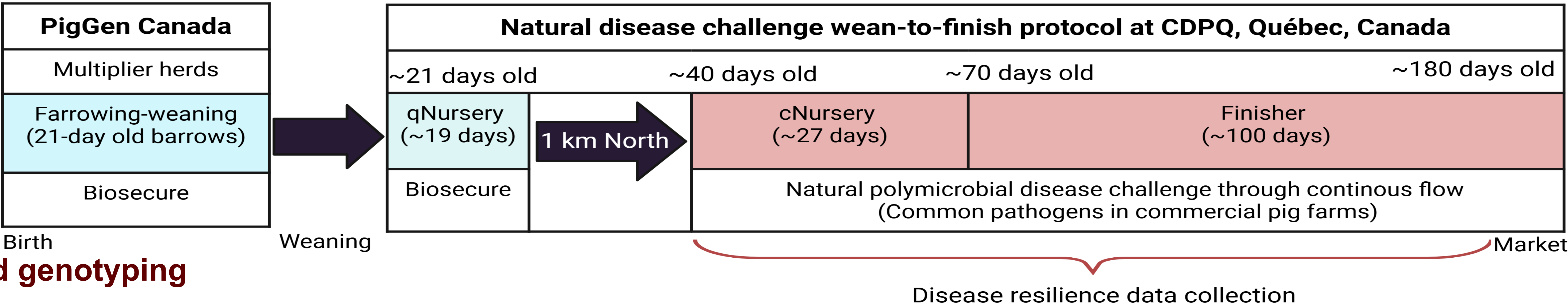
- ❑ Selection in most pig breeding programs is conducted in high-health nucleus herds, while pigs typically encounter multiple pathogens in commercial herds.
- ❑ A knowledge gap remains on genetic parameters and genomics of disease-related traits of growing pigs under a polymicrobial disease challenge.

Objective

- ❑ Estimate genetic parameters and identify genomic regions and pathways associated with disease of growing pigs exposed to a natural polymicrobial disease challenge.

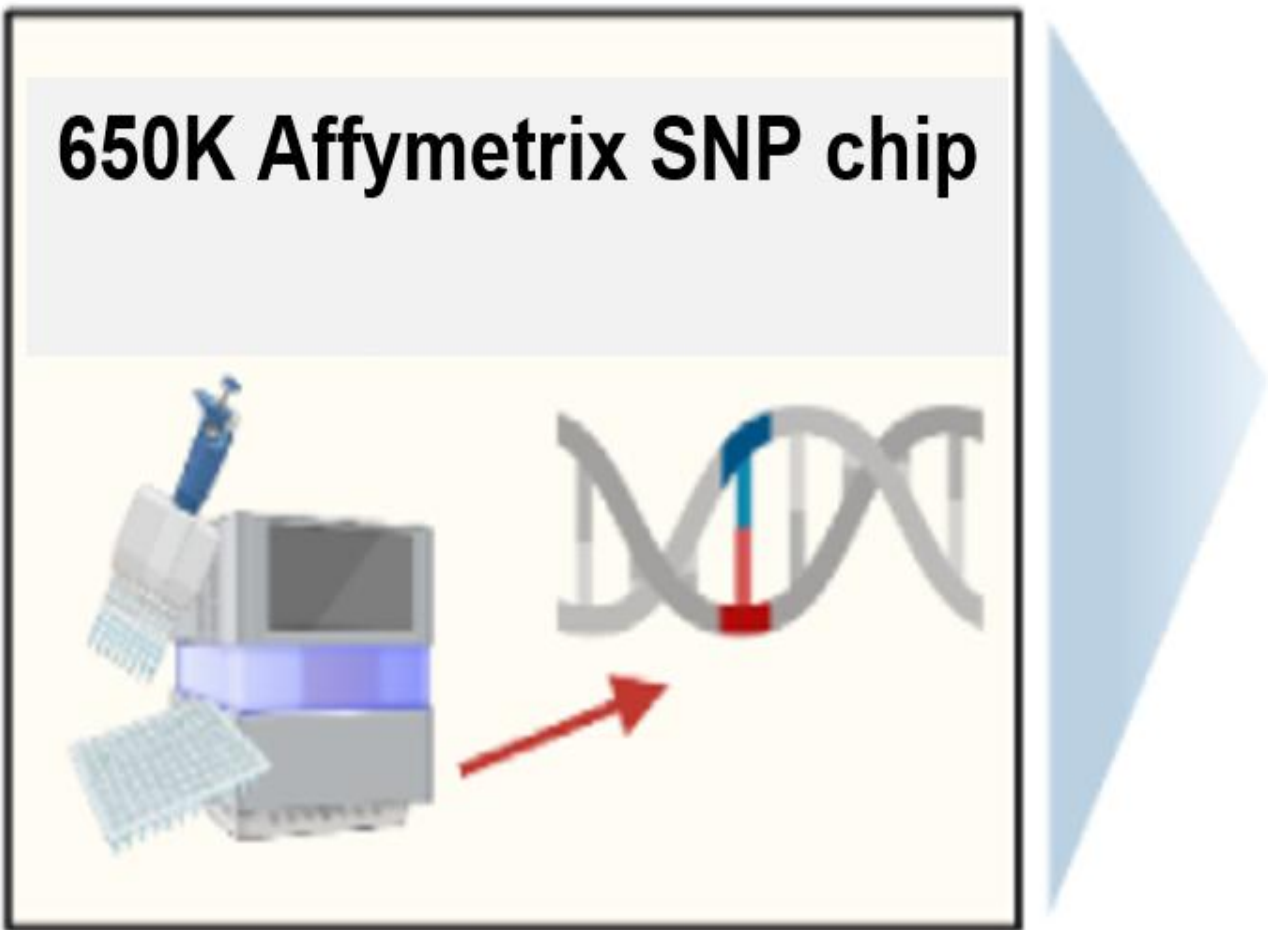


- A three-phase natural polymicrobial disease challenge (Putz et al., 2019, Frontiers in Genetics)



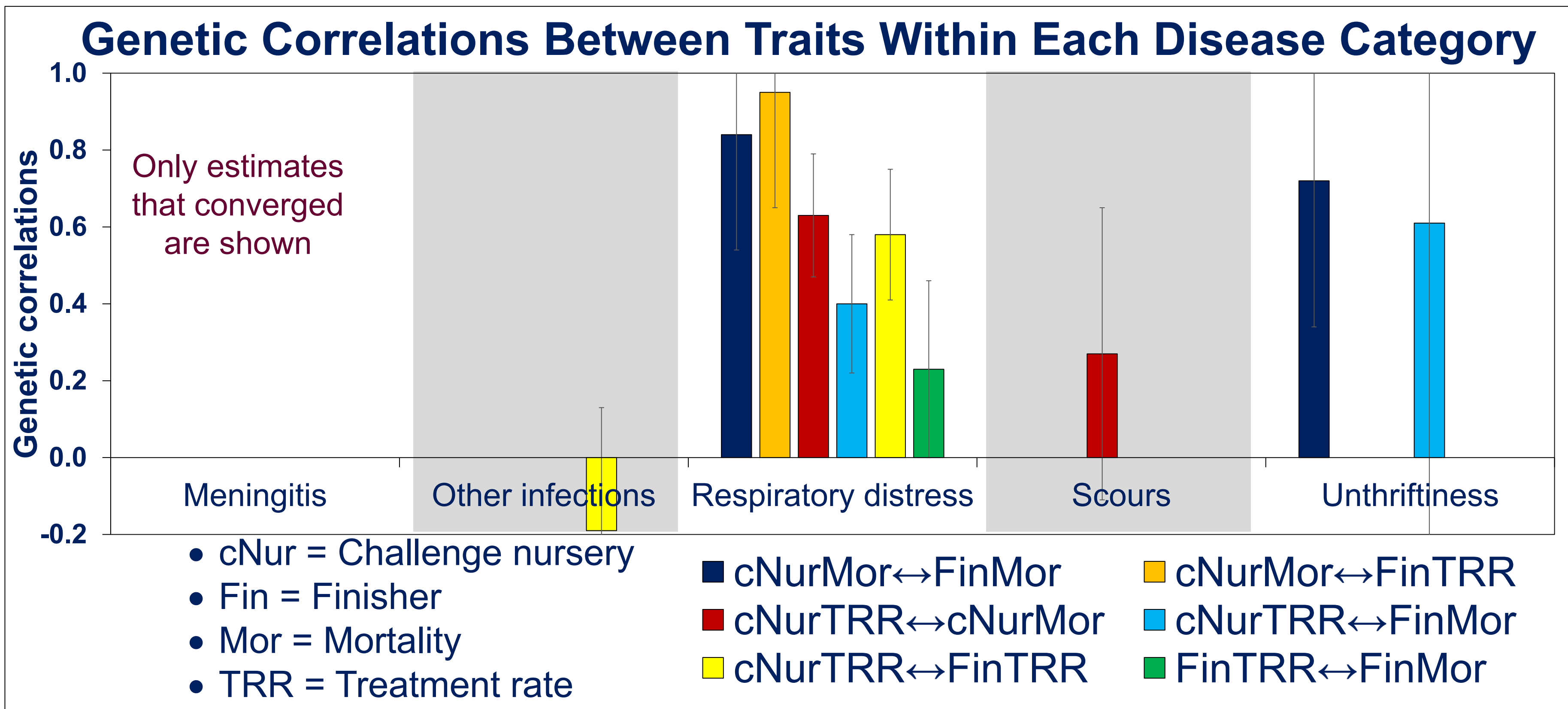
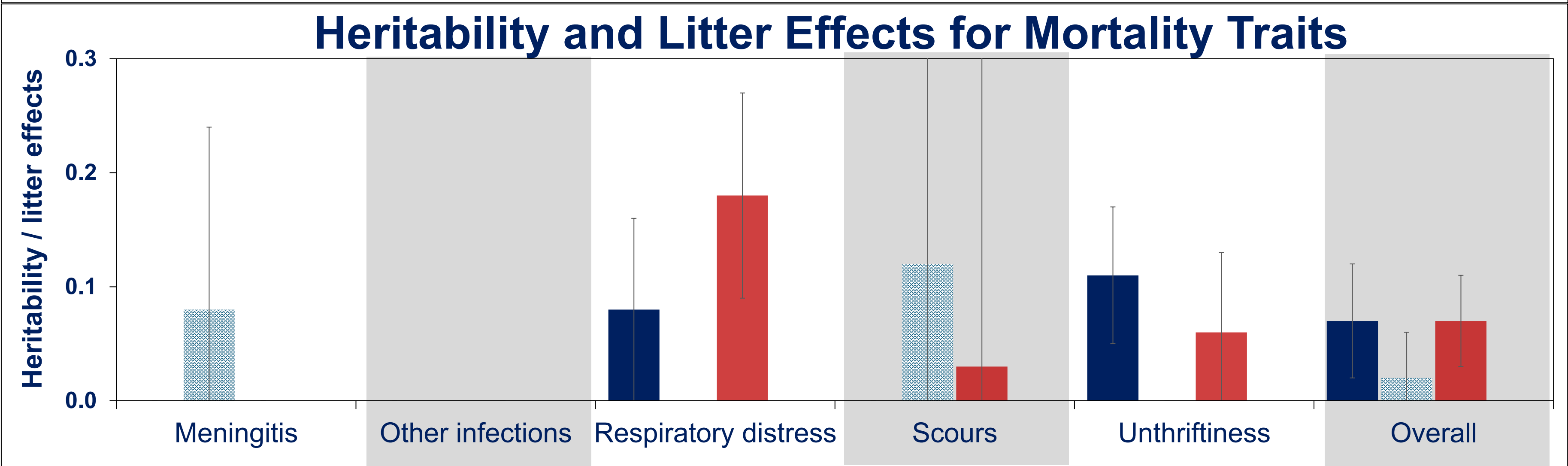
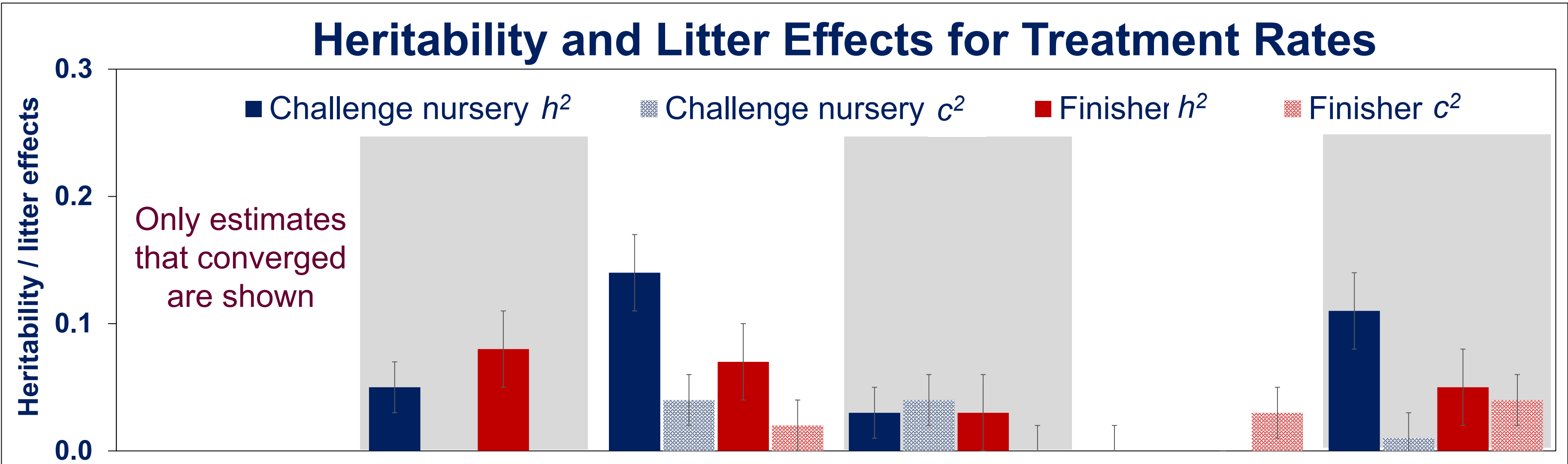
Animals and genotyping

- 4,095 Y × LR barrows
- 65 batches (each ~60/75 pigs)



- Phenotypic data** - Individual health treatments and mortality records categorized as:
1. Meningitis
 2. Respiratory distress
 3. Scours
 4. Unthriftiness
 5. Other infections

- Variance Component Estimation and GWAS:** Uni- and bi-variate analyses using generalized and mixed linear models with genomics (ASReml 4.2); GWAS using JVAS (Cheng et al., 2018).
- Model:** $y_{ijkl} = \text{Batch}_i + \text{EntryAge}_{ijkl} + \text{pen}_j + \text{sow}_{ijk} + \text{pig}(\text{GRM})_{ijkl} + e_{ijkl}$
- Gene Ontology term enrichment analyses:** of 0.25 Mb windows that did not reach significance, using GSEA 4.3.3 (Subramanian et al., 2005).



- ❑ Heritability estimates for treatment rate and mortality were generally higher in the challenge nursery than in the finisher.
- ❑ Compared to other diseases, respiratory distress treatment rate and mortality tended to have higher heritability estimates, both in the nursery and finisher.
- ❑ Litter effects were generally low, but stronger in the challenge nursery.
- ❑ Estimates of genetic correlations between treatment rate and mortality for respiratory distress were generally positive, both in the challenge nursery and finisher.
- ❑ Corresponding genetic correlation estimates for other disease categories were either moderately negative, not positive definite, or did not converge.

CONCLUSIONS

INTRODUCTION

MATERIALS &
METHODS

RESULTS &
DISCUSSION part 1

RESULTS &
DISCUSSION part 2

- ❑ Respiratory distress treatment rate and mortality are substantially influenced by genetics, making it a key target for genetic selection.
- ❑ Litter effects are critical in early life stages but diminish as the pigs mature.
- ❑ Selection strategies to improve health treatment and mortality traits should be balanced to avoid unintended consequences.
- ❑ Genetic parameters and genomics provide the roadmap for breeding programs, and this study provides valuable information to breed for disease resilience in pigs.
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